CELESTE HEBERER

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EDUCATION

University of San Francisco (USF), San Francisco, CA — Expected Graduation: May 2027
Double Major: B.S. in Physics, B.S. in General Engineering (Concentration in Sustainable Environments Engineering)

EXPERIENCE

USF Physics/Astronomy Department, San Francisco, CA — August 2026 - Present
Research Assistant

- Building and validating a Trans-Neptunian Object (TNO) detection service through pipeline-development efforts using archival astronomical observations
- Develop a scalable, AI-assisted discovery framework that can combine archival mining, automated source classification, human-in-loop verification and targeted follow-ups for recovering known objects towards the discovery of new sources.

Mortenson, Sparks, NV — June 2026 - August 2026
Field Engineer Intern (Civil Group)

- Supported 2 heavy-civil project sites, splitting time between field and office operations while conducting daily inspections, coordinating subcontractors, and tracking site progress with project schedules.
- Developed Integrated Work Plans (IWPs) for each stage of construction and presented plans to equipment operators, improving communication of site priorities and safety expectations across active crews.
- Collaborated with foremen and superintendents on daily scheduling, site planning, cost estimating, and quantity takeoffs for both projects, providing field and office teams with the information needed to track project needs.
- Served as a primary communication link between operators, subcontractors, and office staff, relaying real-time field conditions, progress updates, and project information to leadership to support timely coordination and decision-making.

USF Engineering/Physics Department, San Francisco, CA — January 2025 - May 2025
Teaching Assistant: Physics For Engineers (ENGR 150)

- Develop, update, and grade weekly homework assignments for 33 students
- Host office hours to guide students through conceptual topics and coursework.
- Attend weekly meetings with the professor to discuss new topics, lecture styles, student development, and potential course adjustments.

PROJECTS

Sustainable Water Monitoring System – Capstone Project — August 2026 - Present

- Developed a deployable sensor system to monitor water flow, humidity, and air/water temperature, enabling researchers and farmers at Star Route Farms to track river conditions and optimize water storage
- Conduct research on Algae Assisted Microbial Fuel Cells (MFC's) which converts chemical energy into bioelectricity through the catalytic activity of living microbes found onsite

Single Photon Quantum Optics — Spring 2026

- Examined correlations between photodetections at transmission and reflection outputs to experimentally demonstrate both classical and quantum mechanical fields of light through coincidence programming of individual photocounts
- Optimized experimental equipment including an entangled photon source, optical components, and three photon-counting silicon photodetectors and applied normalized intensity correlation functions to identify Poissonian and anti-Poissonian photon distributions to compare classical vs. quantum behavior

Reusable Concrete and COB Molds — Spring 2026

- Designed reusable and sustainable molds for COB and concrete fabrication processes to reduce material waste and prevent damage to samples during testing
- Developed 9 concrete mold designs with 5 lining options, including dissolvable, reusable, and removable materials

SKILLS

- **Technical**: Autodesk (Fusion, AutoCAD, Revit), Soldering, Adobe Illustrator, Arduino IDE, 3D Modeling/Printing, Laser Cutting, Wood Shop, Python (MatPlotlib, Scipy), C++
- **Management:** Microsoft Office, Google Suite, Design Research, Business Marketing & Building

EXTRACURRICULARS

Hispanics in STEM Societies (HISS/SHPE) – *Chapter Founder and President*

- Founded USF chapter; oversee operations, planning, finances, task delegation, and documentation
- Coordinate with national SHPE and plan academic, professional, and cultural events

Society of Women Engineers (SWE) – *Chapter CoFounder*

- Co-founded chapter and supported member recruitment
- Build connections with mentors, affiliates, and grant partners
- Plan and promote events; manage budgeting and fundraising for WE23–25 attendance for 30+ students